Experimental Apparatus

Grasping Force Feedback for da Vinci Surgical Robots

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Our experimental apparatus consists of a custom 3D printed gripper, 3D printed motor and gripper mount, a Hapkit board, a DC motor, a capstan drive with a wire, a force sensor, a pressure sensor, a breadboard, resistors, a switch and all the associated wiring.

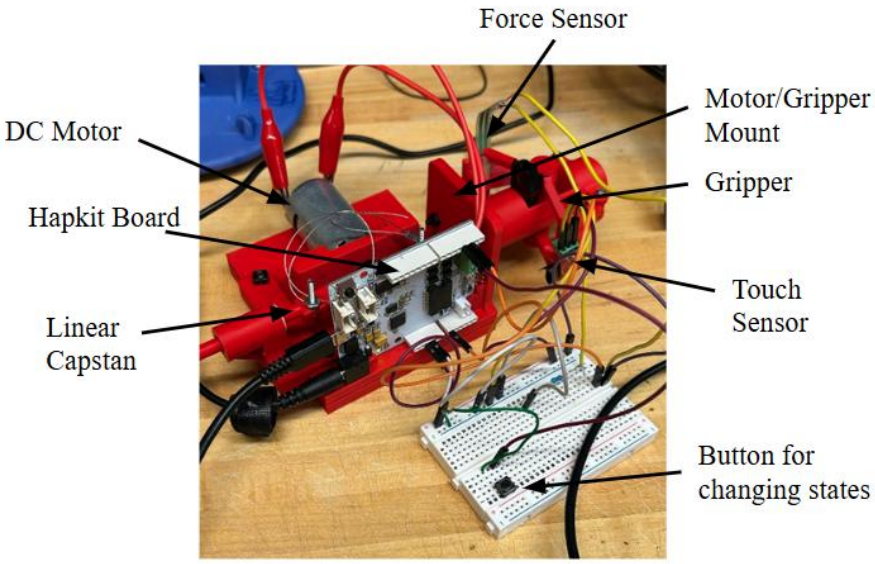


Figure 1. Experimental Apparatus.

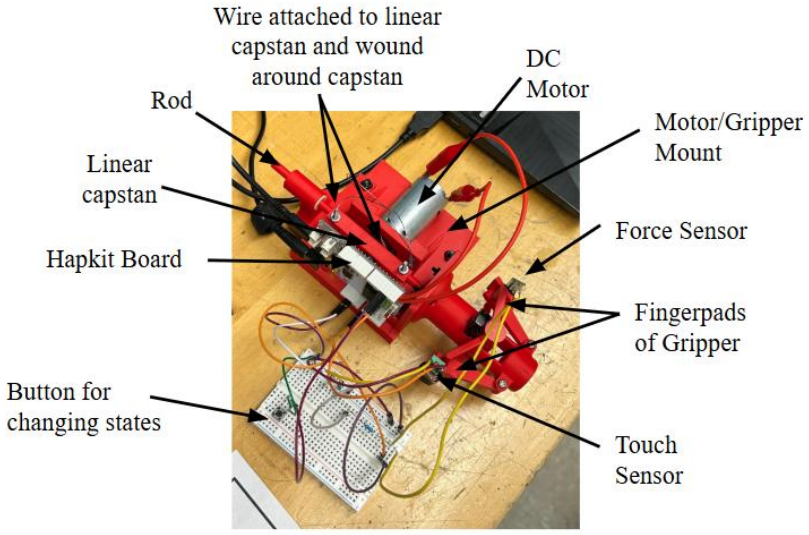


Figure 2. Top view of experimental apparatus.

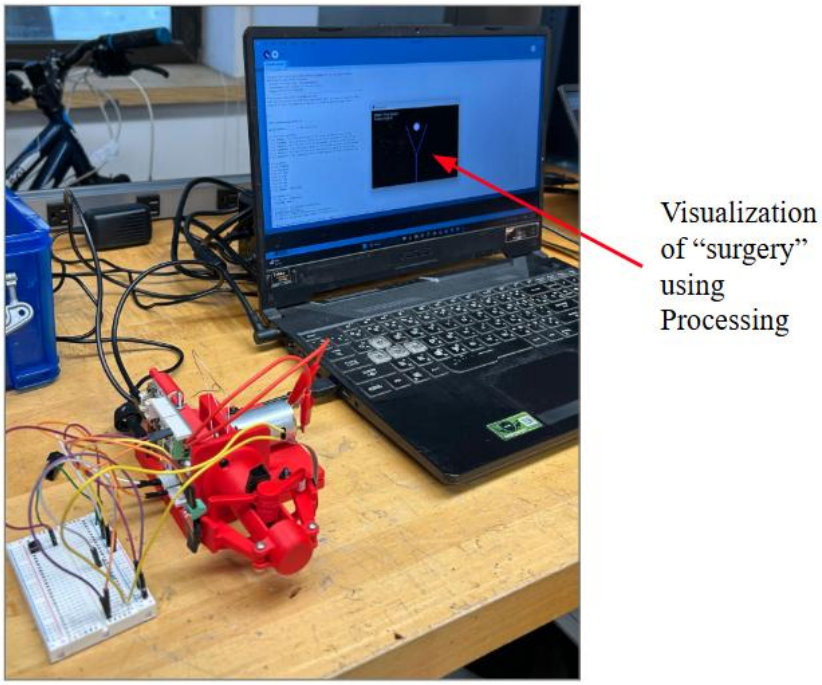


Figure 3. Processing used for visualization of the "surgery".

Non-Significant Risk Justification

This device is non-invasive and only comes into contact with the user's fingertips. Given the simplicity of the mechanical design and the safety features described below, this device poses a non-significant risk.

A. 3D Printed Gripper

The 3D printed gripper provides the point of interaction with the user (see figure 1 and 2). The user applies force to close the gripper, and device applied a controlled force back to the fingertips to simulate grasping. The gripper was printed using PLA.

Safety Features:

- PLA was used to 3D print the gripper, a material safe for contact with skin [1].
- The movement of the grippers is restricted, limiting the fingertip displacement and the maximum force applied, and therefore preventing potential injury.

B. 3D Printed Motor/Gripper Mount

The 3D printed motor/gripper mount is for mounting the motor and the gripper (see figure 1 and 2). It ensures proper alignment and stability of the motor and the gripper.

Safety Features:

- PLA was used to 3D print the mount, a material safe for contact with skin [1].
- The mount remains stationary during operation and sits securely on a flat surface, minimizing any risk of unintended movement.

C. Hapkit Board

This Hapkit board functions similarly to Arduino Uno, but also includes an integrated motor driver and MR sensor (for position sensing of the motor) (see figure 1 and 2). It controls motor actuation, reads motor position, and interfaces with the switch, force sensor, and touch sensor.

Safety features:

- The board operates at low power through either a Micro-USB connection or a small external power supply, reducing the likelihood of generating high forces or hazardous electrical conditions.

D. Motor

The motor is a Micro 6-12V DC Motor (see figure 1 and 2), which drives the capstan mechanism, producing the linear displacement of the rod that generates force feedback at the gripper.

Safety features:

- The motor was selected to ensure that no more than 5 N of force can be transmitted to the user through the gripper.
- The motor has sufficient stall torque to avoid overheating or failure during operation.

E. Capstan drive and wire

The capstan drive converts the motor's rotational motion into linear movement of the rod, which in turn provides force feedback. The wire is secured to both ends of the linear capstan (see figure 2).

Safety features:

- The rod is constrained at both ends, limiting its maximum travel to prevent excessive force output.
- The wire is securely attached and does not interact directly with the user, reducing the likelihood of accidental contact.

F. Force Sensor

The force sensor is mounted on one of the fingerpads of the gripper, and the touch sensor is mounted at the other (see figure 1 and 2). The force sensor measures how much force is being exerted on the fingertip, which is displayed in real time in a Processing simulation. The touch sensor detects when the user is no longer gripping the device and triggers the system to return the gripper to its resting position.

Safety features:

- Both sensors operate at low voltages and pose no electrical hazard.
- The sensors remain cool during operation and do not produce heat that could cause discomfort or injury.

[1] Yang Z, Yin G, Sun S, Xu P. Medical applications and prospects of polylactic acid materials. iScience. 2024 Dec 1;27(12):111512. doi: 10.1016/j.isci.2024.111512. PMID: 39759018; PMCID: PMC11699620.